JINSEONG PARK

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Summary

Affiliation	Korea Institute Advanced Study (KIAS), Seoul, Republic of Korea
	Center for AI and Natural Sciences, AI Research Fellow (Mar 2025 - Present)
Research Topics	AI Data and Model Generalization for SAFE and Temporal Data

Field of Interest

SAFE AI

- Differentially private training to mitigate privacy-utility trade-offs.
  - [C6] investigates a framework that leverages recent diffusion models to amplify the information in limited data for differentially private training.
  - [C3] investigates the strength of flat minima for gradient perturbation and proposes a new sharpness-aware optimization method designed for differentially private deep learning.
  - [J2] proposes a privacy-preserving kernel support model using dynamical systems.
- Generalization to mitigate the performance gap between training and test accuracies.
  - [C2] analyzes the Hessian matrix of neural network parameters and investigates the Jacobian subspace for generalization.
  - [C4] presents a large-scale analysis of robust overfitting to find effective measures for robust generalization against adversarial attacks.
- Adversarial defenses against various attacks to address AI safety.
  - [C1] improves certifiable training against adversarial attacks by analyzing the smoothness of the loss landscape.
  - [J3, J1] analyze the transferability of audio adversarial examples and present transferable adversarial attacks and detection methods by injecting additive noise.
- Relevant Topics
  - [C5] analyzes the fair sampling of diffusion models to generate obfuscated sensitive attributes in synthetic data without relying on classifiers.

Time Series Analysis

- Generative diffusion models to simulate temporal time series data.
  - [P3] designs a diffusion bridge for better diffusion prior selection in time series synthesis models.
  - [J5] enhances financial simulation with synthetic images of price charts by training diffusion models on financial price chart images.
- Forecasting and real-world applications, such as in finance.
  - [C7] proposes an efficient architecture for long-range time series forecasting by eliminating self-attention and leveraging cross-attention mechanisms instead.
  - [J4] proposes a time-efficient optimization method for time series classification and forecasting by capturing the trend and seasonal components of time series data.

Note: [C#] and [J#] denote Conference and Journal publication numbers, respectively.

Education

Mar 2022 - Feb 2025 Ph.D. in Industrial Engineering, Seoul National University
- [Thesis] Generalization Frameworks for Temporal and Private Deep Learning: From Synthetic Data Generation to Robust Optimization
- Supervisor: Jaewook Lee
- Cumulative Major GPA: 3.94/4.30

Mar 2020 - Feb 2022 Master of Science in Industrial Engineering, Seoul National University
- [Thesis] Differentially Private Multi-class Classification Using Kernel Supports and Equilibrium Points
- Supervisor: Jaewook Lee
- Cumulative Major GPA: 4.19/4.30

Mar 2019 - Aug 2019 Bachelor's programme in Mechanical Engineering (Student exchange programme), Technische Universität Graz in Austria

Mar 2016 - Feb 2020 Bachelor of Science in Industrial and Management Engineering, Pohang University of Science and Technology (POSTECH)
- Graduated with Honors: Summa Cum Laude (rank: 1/22)
- Cumulative GPA: 3.91/4.30, Major GPA: 4.03/4.30

Publications

† indicates equal contribution.

JOURNALS

[5] **Jinseong Park**, Hyungjin Ko, Jaewook Lee, "Modeling Asset Price Process: An Approach for Imaging Price Chart with Generative Diffusion Models". In: *Computational Economics* (2024), pp. 1–27.

[4] **Jinseong Park**, Hoki Kim, Yujin Choi, Woojin Lee, Jaewook Lee, "Fast sharpness-aware training for periodic time series classification and forecasting". In: *Applied Soft Computing* (2023), p. 110467.

[3] Hoki Kim, **Jinseong Park**, Jaewook Lee, "Generating Transferable Adversarial Examples for Speech Classification". In: *Pattern Recognition* 137 (2023), p. 109286.

[2] **Jinseong Park**, Yujin Choi, Junyoung Byun, Jaewook Lee, Saerom Park, "Efficient differentially private kernel support vector classifier for multi-class classification". In: *Information Sciences* 619 (2023), pp. 889–907.

[1] Yujin Choi, **Jinseong Park**, Jaewook Lee, Hoki Kim, "Exploring Diverse Feature Extractions for Adversarial Audio Detection". In: *IEEE Access* 11 (2023), pp. 2351–2360.

CONFERENCES

[7] Dongbin Kim, **Jinseong Park**, Jaewook Lee, Hoki Kim, "Are Self-Attentions Effective for Time Series Forecasting?" In: *Advances in Neural Information Processing Systems* (2024).

[6] **Jinseong Park**†, Yujin Choi†, Jaewook Lee, "In-distribution Public Data Synthesis with Diffusion Models for Differentially Private Image Classification". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2024).

[5] Yujin Choi†, **Jinseong Park**†, Hoki Kim, Jaewook Lee, Saerom Park, "Fair Sampling in Diffusion Models through Switching Mechanism". In: *Proceedings of the AAAI Conference on Artificial Intelligence* (2024).

[4] Hoki Kim, **Jinseong Park**, Yujin Choi, Jaewook Lee, "Fantastic Robustness Measures: The Secrets of Robust Generalization". In: *Advances in Neural Information Processing Systems* (2023).

[3] **Jinseong Park**, Hoki Kim, Yujin Choi, Jaewook Lee, "Differentially Private Sharpness-Aware Training". In: *International Conference on Machine Learning* (2023).

[2] Sungyoon Lee, **Jinseong Park**, Jaewook Lee, "Implicit Jacobian regularization weighted with impurity of probability output". In: *International Conference on Machine Learning* (2023).

[1] Sungyoon Lee, Woojin Lee, **Jinseong Park**, Jaewook Lee, "Towards Better Understanding of Training Certifiably Robust Models against Adversarial Examples". In: *Advances in Neural Information Processing Systems* 34 (2021), pp. 953–964.

WORKSHOPS

[2] Seungju Lee, **Jinseong Park**, Jaewook Lee, "Enhancing non-linear asset volatility forecasting models with investor sentiment and explainable AI". In: *Korea Computer Congress (KCC) 2023, XAI workshop, Best paper awards (2/37)* (2023).

[1] **Jinseong Park**, Jaewook Lee, "Wine Recommendation System using BERT". In: *Spring Conference of Korean Institute of Industrial Engineers (KIIE)* (2021).

PREPRINTS

[4] Additional Private Under-review papers.

[3] **Jinseong Park**†, Seungyun Lee†, Woojin Jeong, Yujin Choi, Jaewook Lee, "Leveraging Priors via Diffusion Bridge for Time Series Generation". In: *arXiv preprint arXiv:2408.06672* (2024). Submitted.

[2] Hoki Kim, **Jinseong Park**, Yujin Choi, Jaewook Lee, "Stability Analysis of Sharpness-Aware Minimization". In: *arXiv preprint arXiv:2301.06308* (2023). Submitted to IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI).

[1] Hoki Kim, **Jinseong Park**, Yujin Choi, Woojin Lee, Jaewook Lee, "Exploring the Effect of Multi-step Ascent in Sharpness-Aware Minimization". In: *arXiv preprint arXiv:2302.10181* (2023).

Patents

[Patent Application] 10-2023-0087024, Korea Jun 2023

- 적대적 음성 예제를 감지 방법 및 시스템 (METHOD AND SYSTEM FOR DETECTING ADVERSARIAL SPEECHEXAMPLE)

[Patent Application] 10-2023-0087006, Korea Jul 2023

- 적대적 음성 예제 생성 방법 및 시스템 (METHOD AND SYSTEM FOR GENERATING ADVERSARIAL SPEECHEXAMPLE)

[Patent Application] 10-2023-0073178, Korea Jul 2023

- 커널 서포트 벡터에 기반한 차분 프라이버시 다중 분류 학습을 위한 장치 및 방법 (Apparatus and Method for Learning Differential Privacy Multi-Classification Based on Kernel Support Vector)

Project/Experiences

[Project] COSMAX, Seoul, Korea Dec 2023 - Feb 2024, Jun 2024 - Sep 2024

- Prediction and Generative Neural Networks for Color Cosmetics

[Intern] Industrial Data Engineering & Analytics Lab, Pohang, Korea Sep 2019 - Dec 2019

- Research Intern, hosted by Prof. Hyunbo Cho at POSTECH

[Intern] SK Telecom, Seoul, Korea Jun 2018 - Aug 2018

- R&D Digital Transformation Team Intern
- Developed services for R&D Digital Transformation of internal data clusters

[Project] LNG industry research project with POSCO Energy, Pohang, Korea 2018

- Researcher, with Prof. Ribin Seo at POSTECH
- Organized LNG value chains and strategic implications of POSCO Energy

Teaching Experiences

[Lecturer] Soongsil University, Seoul, Korea 2024

- Analysis of Financial Time Series (2024 Fall Semester)

[TA] Teaching Assistant for Prof. Jaewook Lee, SNU 2020 - 2023

- Undergraduate courses: Statistics for Industrial Engineering (Spring 2020, Spring 2022, and Spring 2023), Mathematical Methods for Industrial and Management Engineering (Fall 2020)
- Graduate courses: Data Mining Technology (Fall 2022), Advanced Topics in Statistical Learning (Spring 2023)

[Instructor, TA] Education Programs, SNU 2020 - 2023

- Instructor for Big data analysis with Python and deep learning with PyTorch in Woori Bank (2022)
- Instructor for AI practice course in Korea Institute of Startup & Entrepreneurship Development (KISED) (2022)
- TA for Machine Learning and Deep Learning for Prof. Jaewook Lee in SNU Big-Data FinTech course (2021 and 2023)
- TA for Python programming for Prof. Namhyung Kim in SNU Big-Data FinTech course (2022)
- TA for Optimization and Linear Algebra for Prof. Jaewook Lee in Samsung Data Scientist for Device Solution (2020-2022) and Industrial Bank of Korea (2020 and 2021)

Valuable Experiences

Invited Talks

- Deep learning with differential privacy; Desilo (Seoul, Korea) Jul 2024

External Courses

- Algorithmic Fairness course in Graduate Summer School (LA, US)
  Institute for Pure & Applied Mathematics (IPAM) at UCLA;
  Organized by Cynthia Dwork, Guy Rothblum, and Noa Dagan Jul 2022

Academic Reviewers

- Journal: Pattern recognition
- Conference: CVPR 2024, NeurIPS 2024

Scholarship

2022 - 2023	Teaching Assistance scholarship, Seoul National University, Full tuition for all semesters.
2021 Fall	Scholarship for Academic Excellence, Seoul National University, Merit-based, 30% of tuition for a semester.
2019 Spring	Scholarship for Exchange Students, POSTECH.
2018 Fall, 2019 Fall	Mentor Scholarship for IME Information System Tech, C++ programming, POSTECH.
2018 - 2019	National Scholarship for Science and Engineering, Republic of Korea, Merit-based, full tuition for all semesters.
2016 - 2017	Scholarship for Academic Excellence, POSTECH, Merit-based, full tuition for all semesters.

Skills

Programming	Python (PyTorch for AI), C++, C, LaTeX	
Leadership	Manager of Association of POSTECH Grown Companies (APGC)-Lab, POSTECH Start-up Incubator	(2018-2019)
	Member and Honorary Member of POSTECH Broadcasting System (PBS)	(2016-2018)
Language	Korean (Native), English (Advanced)	